

Solutions to Quiz 5

Question 1

Is the polynomial $X^3 - 24X - 3$ irreducible? Justify your answer.

Solution:

We note that this is a polynomial with integer coefficients and content is 1.

Thus, if this polynomial is factorisable, we have

$$P(X) = X^3 - 24X - 3 = (X + a)(X^2 + bX + c)$$

where a, b, c are *integers*.

Thus means that $ac = 3$, so that a divides 3.

This means that some element of the set $\{\pm 1, \pm 3\}$ is a root of this polynomial.

We easily check that *none* of these values is a root of the polynomial. It follows that the polynomial is irreducible.

Question 2

Given that $X^4 - X^2 - aX - 1$ has a quadratic factor in $\mathbb{Q}[X]$ for some integer value a , give a possible value for a .

Solution:

Since this is a polynomial with integer coefficients and content 1, we must have

$$Q(X) = X^4 - X^2 - aX - 1 = (X^2 + bX + c)(X^2 + dX + e)$$

for integers b, c, d, e .

Equating coefficients, we obtain the equations

$$\begin{aligned} 0 &= b + d \\ -1 &= bd + c + e \\ -a &= be + cd \\ -1 &= ce \end{aligned}$$

The last equation means that $c = \pm 1 = -e$. Then the second equation gives us $bd = -1$ so that $b = \pm 1 = -d$. From this, we see that $a = \pm 2$.

Alternatively, we note that $X^2 + aX + 1$ is $(X + a/2)^2$ for $a = \pm 2$. It follows that $Q(X) = (X^2 + X + a/2)(X^2 - X - a/2)$ for these values of a .